

Highlights

Intrinsics4Tun: proxy-guided stabilisation and guarded reflection for label-free tunnel-lining segmentation

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- Deployment-time tunnel-lining segmentation framed as label-free accept/abstain control using only pipeline-internal evidence.
- Physical-label instability decomposed into four named layers: seeded parameters, gravity/K anchoring, branch-order switching, and guarded K-frame reflection.
- Absolute proxy used only to choose between plus/minus order hypotheses; relative proxy used only to gate per-ring accept/abstain on K-reflection candidates.
- Stabilisation contribution on 40 held-out K-bearing rings: rotation-ambiguous baseline 0.1445 $\rightarrow$ K-anchored default 0.2554 $\rightarrow$ stabilised candidate 0.3957 fixed-class mIoU; cumulative lift +0.251 (paired bootstrap 95% CI [+0.176, +0.327], Wilcoxon $p < 10^{-6}$).
- Guarded relative reflection layer reported as a safety mechanism: 11 evaluable rings, 2 accepts, 9 abstentions, zero accepted degradations under frozen guardrails; no quantitative held-out lift is claimed for this layer.

Intrinsics4Tun: proxy-guided stabilisation and guarded reflection for label-free tunnel-lining segmentation

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ABSTRACT

Automated segmentation of tunnel-lining components from point clouds enables segment-wise inspection measurements such as joint-gap width, leakage area, and deformation, but new tunnels rarely provide ground-truth labels at deployment. A practical system must therefore decide whether each output is physically meaningful using only pipeline-internal evidence. We present **Intrinsics4Tun**, a label-free control framework for deterministic, projection-based ring-level tunnel-lining segmentation. The framework decomposes deployment instability into four ordered ambiguity layers: seeded physical parameters initialise the pipeline; gravity and keystone anchoring stabilise the unfolded frame; an absolute branch-quality proxy chooses between the two deterministic order hypotheses around the keystone to define a stabilised baseline (candidate0); and a relative proxy with guardrails decides per-ring whether a K-frame reflection candidate should override candidate0 or whether the system should abstain. Ground truth is used during design-time calibration and end-of-pipeline audit, but never inside per-ring deployment decisions. Two findings are reported on a 40-ring K-bearing held-out panel from 17 tunnel sections. *Stabilisation is the load-bearing layer*: keystone anchoring lifts mean fixed-class mIoU from a rotation-ambiguous baseline of 0.1445 (95% CI [0.123, 0.170]) to a K-anchored deterministic default of 0.2554 [0.193, 0.325] (paired bootstrap lift +0.111 [+0.036, +0.193], Wilcoxon $p < 10^{-3}$), and the combined stabilisation stack reaches candidate0 mean 0.3957 [0.328, 0.465] (cumulative lift over the rotation-ambiguous baseline +0.251 [+0.176, +0.327], $p < 10^{-6}$). *Guarded relative reflection is a safety layer*: on the 11 rings where K-reflection candidate evidence exists, the relative proxy and structural guardrails accept 2 challengers, abstain to candidate0 on the other 9, and produce zero accepted degradations under the frozen guardrails on the full panel. We do not claim a quantitative held-out lift for the final layer; it is reported as an accept/abstain control mechanism, not as a mIoU improver. The conclusion is bounded to the tested K-bearing segmental-ring setting: calibrated intrinsic proxies and structural guardrails provide a defensible stabilisation step and a safety floor for downstream refinement, but they do not replace ground truth and leave substantial oracle headroom unused on rings the guardrails refuse.

1. Introduction

Segmental tunnel linings are modular, repeating structures. A typical Tunnel Boring Machine (TBM) lining ring is assembled from a fixed set of concrete segments: a keystone (K), the two adjacent segments that flank it, and the remaining base segments. For inspection and asset management this regularity is valuable, because downstream measurements such as joint-gap width, leakage area, radial deviation, and segment-wise deformation become meaningful only after each predicted segment is assigned to its corresponding physical lining component [? ? ?]. The central observation of this paper is that the assignment from regions to physical identities is fragile in a way that visual segmentation quality does not reveal: an output can partition a ring into clean, segment-shaped regions and still be physically incorrect because the regions are anchored to the wrong segment identities.

At deployment, a tunnel-lining segmentation system must generalise across many tunnels without per-site retuning and decide whether each output should be accepted, refined, or flagged *without* ground-truth labels for the current ring. Existing approaches do not address this requirement directly. Expert-designed pipelines [? ? ?] are interpretable but their performance depends on parameters tied to point density, projection quality, boundary contrast, ring layout, and keystone cues [? ? ?]. Supervised deep-learning segmenters [? ? ? ? ?] can perform strongly when training data match deployment conditions but produce miscalibrated labels under distribution shift, without an explanation of the failure. Label-free quality-estimation methods developed in medical imaging and general vision [? ? ?] typically assume a learned probabilistic segmenter or a labelled reference set retained at deployment; a deterministic, projection-based tunnel pipeline provides neither.

We study deployment-time, label-free control for such pipelines by scoring candidate segmentations with internal pipeline signals: depth-map projections,

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boundary detections, mask completions, layout statistics, and keystone evidence. The experimental pipeline projects each ring to a panoramic depth map, detects segment boundaries in image space, and reprojects the resulting boundary-defined segments to the point cloud [?]. Two design choices set the framework apart from earlier projection-and-prompt segmenters. First, evidence at deployment is taken only from the pipeline’s own intermediate artefacts, which are produced for every ring without labels. Second, the choice of which intermediate signals to trust as quality indicators is treated as a design-time problem, because a metric that rewards visual regularity can be fully satisfied by a physically wrong segmentation.

Calibrated diagnostics rather than hand-picked heuristics. Internal metrics are not fixed from domain priors. We run Bayesian optimisation *offline* on a small set of labelled calibration rings that span distinct point-density, spatial-distribution, and keystone configurations [?]. The BO trajectory is mined for two artefacts that an optimiser usually discards: a per-parameter sensitivity profile and a per-metric quality-correlation profile. The selected parameters, proxies, and guardrails are frozen before held-out evaluation. A small labelled set therefore supports label-free decisions on rings the calibration never saw.

A structural failure mode of intrinsic self-assessment. Unfolding a ring to a panoramic depth map requires choosing a starting angle. That choice rotates the entire projection, but spacing uniformity, boundary continuity, mask completeness, and similar layout-regularity metrics are all invariant to the rotation. A segmentation can therefore satisfy every internal metric, look visually clean, and still be off by one or more segment positions in physical identity (Figure 1). Layout regularity rewards orderly patterns; it does not constrain physical identity. We therefore formulate physical-label alignment as a layered ambiguity-resolution problem with four named layers: seeded parameters establish a deterministic prior, gravity and K anchoring define the frame, an absolute proxy resolves the binary plus/minus order ambiguity around K to define candidate0, and a relative proxy with guardrails tests whether a reflected K-frame candidate is intrinsically better than candidate0.

The two threads are connected. Without deterministic stabilisation, a label-free self-assessment objective is satisfied by an entire rotation orbit of physically distinct segmentations. With the four layers in place, the predictor’s label scheme is tied to a physically consistent frame before any downstream use. Earlier projection-and-prompt pipelines [?] assume a tunnel-direction convention but do not formulate the ambiguity baseline explicitly or evaluate proxy-guided accept/abstain control on a held-out panel.

Contributions. The paper makes three contributions, each tied to a specific block of evidence presented later.

1. **BO as a design-time metric selector.** Bayesian optimisation is run offline on a six-ring calibration panel and its trajectory is mined to identify five sensitive preprocessing parameters and six quality-correlated internal metrics under pre-registered selection rules. Both sets are frozen before held-out evaluation (Section 3.4.5, Table 5–6).
2. **A layered ambiguity-resolution architecture.** We characterise rotational ambiguity in unfolded ring projections as a structural blind spot of layout-regularity metrics (Figure 1) and decompose deployment instability into seeded parameters, gravity/K anchoring, absolute-proxy branch selection, and relative-proxy guarded reflection (Figure 3).
3. **A stabilisation contribution with statistically bounded evidence and a guarded safety layer.** On 40 held-out K-bearing rings, keystone anchoring lifts mean fixed-class mIoU from a rotation-ambiguous baseline of 0.1445 to a K-anchored default of 0.2554 (paired bootstrap lift +0.111 [+0.036, +0.193], Wilcoxon $p < 10^{-3}$), and the combined stabilisation stack reaches candidate0 mean 0.3957 (cumulative lift +0.251 [+0.176, +0.327], $p < 10^{-6}$; Table 2). The guarded relative-reflection layer is reported as an accept/abstain safety mechanism: 11 rings carry K-reflection candidate evidence, the relative proxy and structural guardrails accept 2 challengers, abstain to candidate0 on the other 9, and produce zero accepted degradations on the panel. We do not make a quantitative lift claim for this layer.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the Intrinsics4Tun framework, dataset, and evaluation protocol, including the statistical-design and Polanyi-paradox checks required by the AIC review rubric. Section 4 reports stage-decomposed results, stratified panels, and calibration sensitivity. Section 5 interprets the findings against prior work and bounds the conclusions by limitations and evidence quality. Section 6 concludes.

2. Related work

2.1. Deployment-time quality control for tunnel-lining segmentation

Tunnel-lining segmentation is conventionally evaluated offline against labelled ground truth using metrics such as overall accuracy (OA) or mean intersection-over-union (mIoU) [?]. At deployment, however,

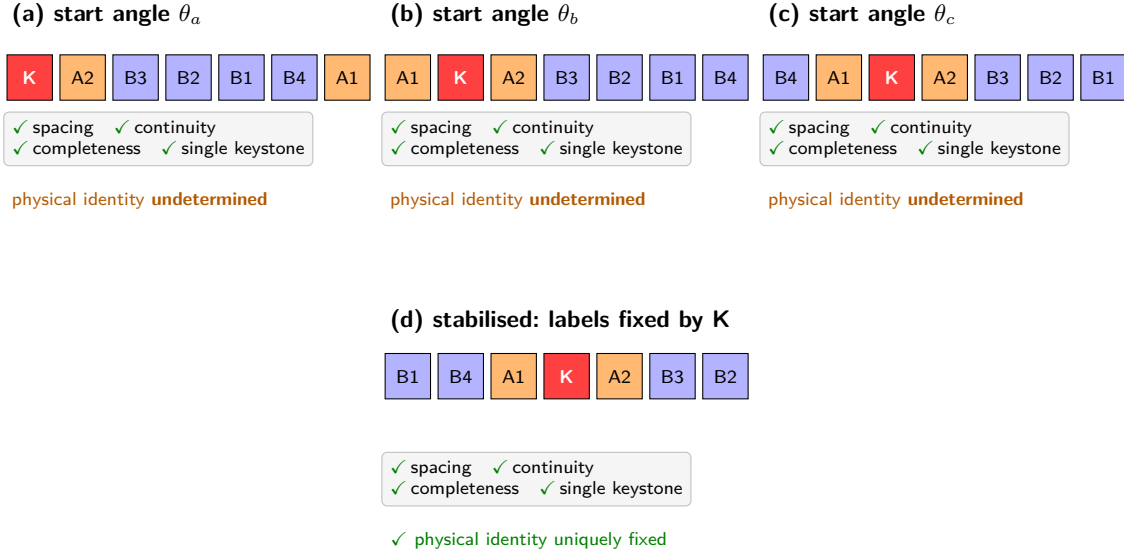


Figure 1: Rotational ambiguity in unfolded ring projections is invisible to layout-regularity self-assessment. (a)–(c) The same physical ring, unfolded with three different starting angles θ . All three projections satisfy every internal metric (spacing uniformity, boundary continuity, mask completeness, single-keystone presence) and produce visually clean segmentations, but the mapping from image columns to physical lining components depends on θ , which is not directly observable from the image. (d) The proposed labelling layer fixes this ambiguity: the keystone is localised first, then all remaining blocks receive canonical class identifiers by signed offset from the keystone. Cell colours and labels are illustrative schematics, not actual depth-map renders.

new tunnels rarely have ground-truth labels, so accept/refine/flag decisions must be made from evidence the pipeline produces itself.

Existing approaches form two families. The first is expert-designed pipelines (e.g., geometric, projection-based, and template-based), whose assumptions and intermediate artefacts can be inspected [? ?]. Their main limitation is brittleness: performance depends on pre-processing parameters tied to point density, projection quality, boundary contrast, ring layout, and keystone cues, so engineers must diagnose failures and retune the pipeline whenever conditions shift [? ?]. The second is supervised deep-learning segmenters, including point-cloud networks and tunnel-specific architectures [? ? ? ?], which can perform well when training data match deployment conditions, but offer no explanation when out-of-distribution geometry, scanning settings, or tunnel layout produce mislabelled segments. Neither family provides an explicit deployment-time mechanism that decides whether an output is plausible enough to accept, fixable enough to refine, or unreliable enough to flag.

2.2. Label-free segmentation quality estimation

Label-free (or test-time) quality estimation in medical imaging and general vision infers segmentation reliability without ground-truth labels for the current case [? ?]. Representative strategies include learned error or quality predictors [? ? ?]; reverse classification accuracy, which reuses labelled reference cases

at assessment time [? ?]; uncertainty calibration for probabilistic segmenters via Bayesian approximations, MC dropout, or deep ensembles [? ? ? ?]; and disagreement-based failure detection from multiple models or auxiliary streams [?]. These methods are effective in their native settings but each assumes resources that a deterministic, projection-based tunnel pipeline does not provide: a learned probabilistic segmenter, a labelled reference set retained at deployment, or multiple model streams. The unresolved gap is therefore not the absence of label-free quality estimation in general but the absence of a mechanism for deterministic pipelines whose only deployment-time evidence consists of intermediate artefacts.

For such pipelines the available evidence is *pipeline-internal*: signals from depth-map projection usability, boundary-defined segment completeness, ring-layout validity, single-keystone presence, assignment stability, and sensitivity to local parameter perturbations [? ?]. We refer to these measurable signals as *internal metrics*, and to a calibrated subset used at deployment for accept/refine/flag decisions as *diagnostics*. Internal metrics cannot replace ground-truth mIoU; they are the only consistent source of label-free evidence in this regime. The open question is which internal metrics genuinely track segmentation quality, and how that question can be answered with the small labelled budget realistic for tunnel projects.

2.3. Bayesian optimisation as a design-time metric selector

Bayesian optimisation is the standard tool when the objective is expensive, noisy, or unavailable in closed form [? ?]. It builds a probabilistic surrogate of the parameter–performance map, typically with a Gaussian process [?], and an acquisition function trades exploitation against exploration [?]. BO is widely applied to parameterised engineering pipelines whose runs are costly and whose parameters interact non-locally; in tunnel segmentation, for example, unfolding and denoising settings affect depth-map coverage and boundary contrast, which propagate into downstream boundary detection and mask prompting [?].

Most uses of BO in segmentation pipelines treat it as a hyperparameter optimiser whose output is a single high-performing configuration. We use BO differently. By recording the parameter–quality trajectory across a small labelled calibration set, BO produces two artefacts that are usually discarded: a sensitivity profile over parameters, and a correlation profile between candidate internal metrics and ground-truth quality. In our framework these are the primary outputs — the parameter set to expose to deployment-time refinement, and the metric set to use as diagnostics. Once selected, both are frozen; deployment is label-free and BO is not rerun.

2.4. Label-free control and where internal diagnostics differ

Once diagnostics are available the system needs a mechanism for responding to them. Label-free quality-control methods ask whether an output can be trusted without ground truth, while self-refinement methods ask how a system can revise an output after receiving feedback [? ?]. For deterministic tunnel segmentation the more basic requirement comes first: define which internal signals are credible evidence, and ensure the predicted label frame is physically meaningful before any refinement is attempted.

These frameworks supply a useful structural template, but they do not specify what should be optimised when no ground-truth signal is available, nor how quality signals should be constrained inside a physical pipeline. The diagnostic and stabilisation layers that must exist before deployment-time refinement is meaningful are: (i) a feedback signal that is a fixed, design-time-calibrated set of internal metrics rather than free-form critique, (ii) a candidate label frame stabilised by keystone-anchored canonical labelling, and (iii) structural plausibility checks that decide when a ring should be accepted for downstream use or flagged as outside the current evidence base.

Positioning. This paper studies deployment-time, label-free control for deterministic, projection-based tunnel-lining segmentation. We inherit the family of label-free quality-estimation *ideas* from the medical-imaging literature but not its assumptions on probabilistic segmenters or labelled reference cases, and we use BO not as a deployment-time optimiser but as a design-time selector of parameters and diagnostic metrics. Empirically, we identify a structural failure mode of intrinsic self-assessment in unfolded projections (rotational ambiguity invisible to layout-regularity metrics) and show that keystone-anchored canonical labelling, combined with absolute-proxy branch selection and guarded relative reflection, can resolve it without ground truth in the tested K-bearing setting. This differs from SAM4Tun-style segmentation [?], which relies on a consistent direction convention but does not formulate the ambiguity baseline explicitly or evaluate a canonical-labelling stabiliser on a held-out panel.

3. Materials and methods

3.1. Methodological overview

Intrinsics4Tun is a deterministic ring-level segmentation pipeline wrapped by a label-free control architecture (Figure 2, Figure 3). The architecture separates *stabilisation*, which produces a physically meaningful baseline (candidate0), from *reflection*, which may only challenge candidate0 under guardrails. The two intrinsic proxies are scoped to disjoint decisions: an *absolute* proxy chooses between the two deterministic plus/minus branches around the keystone, and a *relative* proxy asks whether a K-frame reflection candidate is intrinsically better than candidate0 by a margin that exceeds the calibration threshold.

3.2. Task definition

The unit of analysis is a single K-bearing segmental TBM lining ring represented as a labelled 3D point cloud. The pipeline input is the raw point cloud; the output is a per-point label drawn from the canonical class set $\mathcal{C} = \{\text{background}, K, A_1, A_2, B_1, B_2, B_3, B_4\}$, where $A_{1,2}$ are the two segments adjacent to the keystone and $B_{1...4}$ are the remaining base segments. The deployment-time decision required of the pipeline is, for each ring, to emit either (a) a labelled output assigned to a single deterministic candidate or (b) an abstention back to the stabilised candidate0; in both cases the label scheme must be tied to a physically consistent frame so that downstream measurements (joint-gap width, leakage area, segment-wise deformation) refer to the correct physical components [? ? ?]. Ground truth is read only for design-time calibration and end-of-pipeline audit. The primary outcome variable is fixed-class canonical mIoU on the held-out panel (Section 3.4.3); the practical-meaning threshold and the statistical-design plan are stated in Section 3.4.4.

Ring-based segmentation pipeline: 3D point cloud, 2D projection, canonical masks, and label-free decision control

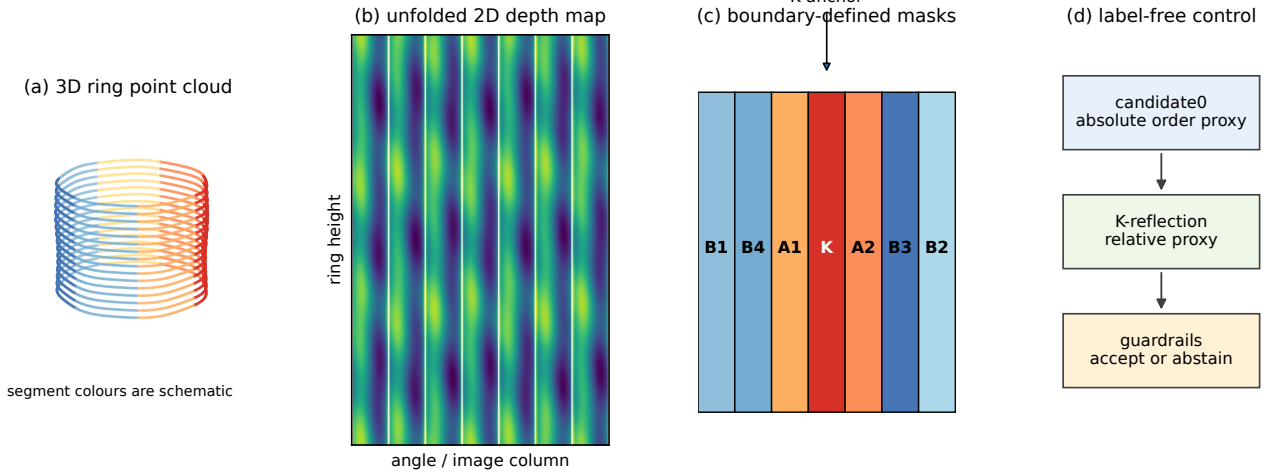


Figure 2: Ring-based pipeline view linking the 3D inspection object, the 2D working representation, and the label-free decision layer. (a) A tunnel-lining ring is a 3D point cloud whose physical components must be recovered as K, adjacent, and base blocks. (b) The deterministic pipeline unfolds the ring into a panoramic depth map so boundary evidence can be detected in image space. (c) Detected boundaries define 2D masks; the masks are useful for asset inspection only after they are tied to a stable physical frame through the K anchor. (d) Intrinsic4Tun adds the deployment-time control layer: candidate0 is produced by stabilisation and absolute-proxy branch selection, while relative-proxy reflection can override candidate0 only when guardrails accept the challenger. The figure is a schematic; quantitative claims are reported in Tables 2–4.

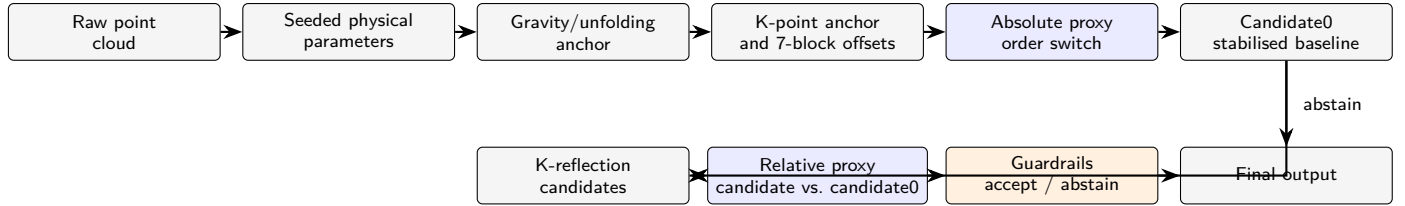


Figure 3: Intrinsic4Tun as layered ambiguity resolution. Seeded physical parameters, gravity/K anchoring, and absolute-proxy order switching define candidate0. K-reflection candidates are evaluated only as challengers to candidate0. The relative proxy and guardrails decide whether to accept the challenger or abstain to candidate0. Ground truth is not used inside the decision path.

3.3. Intrinsic4Tun

The framework has two phases. **Phase 1 (design-time calibration)** runs Bayesian optimisation on a small labelled calibration panel over a parameterised projection-based pipeline. The BO trajectory — not the best-of-trial configuration — is the calibration output: it yields a Spearman sensitivity profile over parameters, a Spearman correlation profile over candidate internal metrics, and, from the upper-quartile-mIoU regime, parameter envelopes and guardrail thresholds. Sensitive parameters, diagnostic intrinsic4s, and guardrail bundles are frozen at the end of Phase 1 (Section 3.4.5).

Phase 2 (held-out label-free deployment) applies the frozen artefacts to each new ring without per-ring labels. The deployment path follows the four-layer ambiguity-resolution structure of Figure 3:

1. *Seeded parameters.* Frozen R4Tun-seed values, clipped to the BO hard upper bound where required, project the ring to a panoramic depth

map. The deployment-time tunable set is restricted to the five preprocessing parameters that cleared the calibration sensitivity threshold (Table 5); all 21 detection parameters are pinned at seed values.

2. *Gravity / K anchoring.* The unfolding origin is chosen by gravity evidence when intrinsic indicators justify it; otherwise the deterministic default applies. The keystone is then localised, and the remaining blocks receive canonical class identifiers by signed offset from K. This step produces two deterministic branch hypotheses around the keystone, denoted *plus* and *minus*.
3. *Absolute-proxy branch selection.* An absolute branch-quality proxy reads visible artefacts (projection usability, boundary structure, layout validity, keystone evidence, branch-specific template/health signals) and selects the plus or minus branch. The chosen branch defines candidate0. The absolute proxy is scoped only to this binary

decision; it is not used to rank K-reflection candidates.

4. *Relative-proxy guarded reflection.* K-reflection candidates are proposed from detected K evidence, horizontal anchors, and physically meaningful local offsets. Each candidate is evaluated by a *relative* proxy whose inputs are candidate-vs-candidate0 deltas in intrinsic features (balance, block-size variance, K ratio, ring completeness, forced-repair count, K-offset plausibility). A challenger overrides candidate0 only if it passes the structural guardrails of Table 7 *and* the relative-proxy score exceeds candidate0 by the calibrated margin $\delta = 0.02$; otherwise the system abstains to candidate0.

Polanyi-paradox check. The pipeline relies on expert tunnel-engineering judgement that is normally used reliably but rarely stated explicitly. We make the boundary between formalised and tacit knowledge explicit here. *Formalised as rules, priors, or constraints:* the canonical seven-block topology (K, A_{1,2}, B_{1...4}), the keystone-anchored offset rule for canonical labelling, the gravity-direction prior, the binary plus/minus order ambiguity around K, and the upper-quartile-mIoU envelopes that define the guardrail thresholds. *Approximated through learned BO trajectories:* the sensitivity profile of preprocessing parameters and the Spearman correlations between internal metrics and ground-truth quality. *Still tacit and not replaced:* the selection of calibration rings (which still encodes an expert choice of regime coverage), the design of the candidate-feature set for the absolute and relative proxies, and the final readout of audit mIoU. The deployment claim is bounded accordingly: the method reduces but does not remove expert dependence; in particular, calibration-panel composition is a human choice that we treat as a limitation rather than an automated step.

3.4. Experimental setup

3.4.1. Dataset

All rings used in this paper are drawn from the Seg2Tunnel benchmark [?]. The unit of analysis is a single ring, labelled at the point level (background, K-block, base blocks, adjacent blocks). The scope of this paper is restricted to K-bearing segmental linings; non-segmental tunnel sections are out of scope. Ground-truth labels are used only at design time — to drive Bayesian optimisation on the calibration panel and to score rings on the held-out panel — and are never supplied to the label-free assessment stage. We define two non-overlapping panels (Table 1): a small calibration panel that BO sees during design, and a larger held-out panel reserved for evaluation under frozen parameters and metrics.

BO calibration panel. The calibration panel contains 6 rings drawn from six distinct tunnel sections, with four rings from the larger-diameter family (4-x

Table 1

Composition of the BO calibration and held-out evaluation panels (K-bearing segmental rings from the Seg2Tunnel benchmark [?]; the two panels do not overlap).

	BO calibration	Held-out
Rings	6	40
Tunnel sections	6	17
Family 4 / Family 5	4 / 2	24 / 16
Density (dense/med./low/sparse)	3/1/1/1	17/13/9/1
Coverage (full / partial)	5 / 1	38 / 2
K-span (narrow/normal/wide)	1/4/1	9/20/11
Joint pattern (canon. / rev.)	3 / 3	20 / 20
Role of GT	BO objective	scoring only

sections) and two from the seven-segment family (5-x sections). The panel is intentionally minimal. Rings are not chosen for coverage but for non-redundant learning value: each ring is selected to expose BO to a different parameter-relevant axis — a dense full-coverage prototype, a density gradient through medium and low tiers, a sparse failure boundary at wide K-span, a narrow K-span case at fixed dense density, and a partial-coverage stress case. Because keystone-anchored canonical labelling at detection time fixes the predictor’s label scheme regardless of the per-ring starting angle (Section 3.3), the keystone-quadrant axis carries no additional information for BO; calibration therefore prioritises density, K-span, joint-pattern, and coverage instead. The parameters and internal metrics that BO surfaces on this panel are frozen and reused at deployment without further tuning.

Held-out evaluation panel. Generalisation is evaluated on a held-out panel of 40 rings drawn from 17 sections, stratified into 22 *cross-section* rings (from sections never touched by BO) and 18 *within-section* rings (from sections that contributed a calibration ring, but in regimes the BO ring did not represent). The two splits answer different transfer questions: cross-section measures whether frozen parameters and diagnostics transfer to entirely unseen sections, while within-section measures whether they transfer across regime shifts (e.g., partial coverage, narrow K-span, sparse density) at the same site. The panel has zero overlap with the calibration set, intentionally retains stress cases (degraded depth maps, sparse density, partial coverage) so that flag-rate behaviour can be measured rather than assumed, and excludes two rings for which the upstream preprocessing fails to produce a usable depth map. At deployment, frozen parameters and diagnostics are applied to each ring without per-ring labels; ground truth is read only afterwards to compute mIoU for reporting.

3.4.2. Experimental design

The held-out comparison is staged to isolate the contribution of each control layer. (i) **Rotation-ambiguous baseline**: image-order assignment without a stable physical frame. (ii) **Stabilised candidate0**: seeded parameters, gravity/K anchoring, physical 7-block offsets, and absolute-proxy order switching. (iii) **Relative-proxy guarded reflection**: K-reflection candidates are generated where candidate evidence is available; the relative proxy and guardrails either accept a challenger or abstain to candidate0. Ground-truth labels are never used to make per-ring branch, gravity, or reflection decisions; they are read only after selection for reporting mIoU.

3.4.3. Evaluation metrics

The primary outcome is fixed-class canonical mIoU on the 40-ring held-out panel. For each ambiguity-resolution layer we report mean mIoU, paired bootstrap 95% confidence interval, lift over the previous layer, number of rings carrying candidate evidence, accept/abstention counts, accepted-degradation count, and missed-oracle opportunity. Hungarian-permutation mIoU is reported as a secondary diagnostic only.

3.4.4. Statistical design

The unit of analysis is a single K-bearing held-out ring; the panel of 40 rings is fixed by stratified design (Section 3.4.1). For each pairwise stage contrast we report the paired difference of per-ring mIoU as the main effect-size estimate. Confidence intervals are constructed by paired non-parametric bootstrap over rings ($B = 10,000$ resamples). We complement the CI with one-sided Wilcoxon signed-rank p -values for lift exceeding zero, but the lead defense of each claim is the estimate together with its CI rather than the p -value alone. We treat a stage contrast as *a defensible lift* when its 95% CI lower bound is strictly above zero and as *not separately defended* when the CI brackets or touches zero, in which case we either combine the layer with its neighbour into a single defensible claim or report the layer descriptively without a lift claim. We treat a layer as *safe on this panel* when the accepted-degradation count is zero under the frozen guardrails; safety is judged separately from lift, because a deployment system can be useful by abstaining even when its lift cannot be statistically separated from zero. Selection rules for sensitive parameters, diagnostic intrinsics, guardrail thresholds, and the relative-proxy margin δ were registered on the BO calibration corpus before the held-out panel was scored.

3.4.5. Sensitivity analysis

We use Bayesian optimisation as a design-time metric selector rather than as a deployment-time hyperparameter optimiser. For each of the six calibration rings we run 30 BO trials over the 13-parameter preprocessing space and 30 trials over the 21-parameter

detection space; each trial executes the full pipeline (Section 3.3) and is scored by fixed-class canonical mIoU against ground truth. Across the panel this yields 180 successful preprocessing trials and 180 successful detection trials, with three preprocessing trials (1.6%) failing with run-time exceptions and excluded from the surrogate model and from all correlations.

Two artefacts of the trial trajectory drive the calibration. First, the pooled Spearman correlation ρ_{Spearman} between each parameter (or each internal metric) and canonical mIoU, computed across all 360 successful trials. Spearman is preferred to Pearson because parameter-mIoU relationships are monotone but not necessarily linear over the BO range. Second, the empirical envelope $[p_{25}, p_{75}]$ of each parameter over the upper-quartile-mIoU regime — the operational range in which segmentation quality is consistent across rings. We apply explicit selection rules so that the choice of tunable parameters and diagnostic metrics is reproducible from the calibration corpus and not retrospectively tuned. A parameter is exposed to deployment-time tuning if its pooled $|\rho_{\text{Spearman}}|$ is at least 0.20 and its parameter-range evidence is present in at least three of six rings; an internal metric becomes a deployment-time diagnostic if its pooled ρ_{Spearman} on the preprocessing-stage corpus is at least 0.50 and the metric is non-null in at least 90% of successful trials. Guardrail thresholds are calibrated to the upper-quartile-mIoU regime: for each diagnostic, the strict cut is the regime p_{25} and the permissive cut is the regime minimum; for the permutation-vs-fixed mIoU gap that motivates anchored unfolding, the strict and operational cuts are the corpus median and p_{75} respectively. The thresholds are calibration-specific and should be re-derived if the calibration panel is replaced.

We then report two complementary forms of evidence for each retained effect. The pooled ρ across 360 trials is the primary point estimate; the per-ring ρ across the six rings is reported as a six-point spread that exposes the sensitivity of each claim to the choice of calibration ring. We treat an effect as high-confidence only when its sign is stable across all six rings; effects whose sign flips on at least one ring are reported as moderate-confidence and discussed as such (Section 4.3). Selection rules and threshold derivations were fixed before the held-out panel (Section 3.4.1) was scored.

4. Results

4.1. Stage decomposition of the held-out result

Table 2 reports the four-layer decomposition of the held-out result. The stabilisation stack (K anchoring through absolute-proxy branch selection) accounts for essentially all of the defensible mean-mIoU gain over the rotation-ambiguous baseline. The final

Table 2

Stage decomposition of the held-out result on 40 K-bearing rings (17 sections). Means and paired bootstrap 95% confidence intervals are computed on per-ring mIoU. Lift columns are paired differences against the previous layer; CIs and Wilcoxon p -values refer to lift > 0 . The combined K-anchored $\rightarrow$ candidate0 row collapses the two intermediate layers because the absolute-proxy-only step alone is not separately defended (its standalone CI brackets zero, see footnote). The final guarded-reflection layer is reported descriptively in this table and as an accept/abstain audit in Table 3; no quantitative held-out lift is claimed for it.

Layer	Mean mIoU	95% CI	Lift vs prev.	Lift 95% CI	Wilcoxon p
Rotation-ambiguous baseline [†]	0.1445	[0.123, 0.170]	–	–	–
K-anchored deterministic plus	0.2554	[0.193, 0.325]	+0.111	[+0.036, +0.193]	$< 10^{-3}$
Stabilised candidate0 [‡]	0.3957	[0.328, 0.465]	+0.140	[+0.056, +0.221]	$< 10^{-3}$
Cumulative defendable stabilisation: baseline $\rightarrow$ candidate0 lift = +0.251 [+0.176, +0.327], $p < 10^{-6}$.					

rotation-ambiguous baseline is a constructed lower bound corresponding to image-order labelling without K anchoring; it is not a deployed comparator. [†] The candidate0 row combines the absolute-proxy branch-selection layer and the subsequent plus/minus refinement, because the absolute-proxy stage examined alone has 95% CI bracketing zero (Wilcoxon $p = 0.06$) and is therefore not separately defended; reporting the two layers as a single combined step is the SOP-compliant alternative. The guarded relative-reflection layer is reported separately in Table 3 as an accept/abstain audit with no quantitative held-out lift claim.

relative-reflection layer is reported separately as an accept/abstain control mechanism (Table 3); the table makes no quantitative lift claim for it.

Stabilisation is the load-bearing layer. The largest defendable contribution is keystone anchoring over the rotation-ambiguous bottom (+0.111, CI [+0.036, +0.193], $p < 10^{-3}$), and the combined absolute-proxy and plus/minus refinement layer adds a further +0.140 (CI [+0.056, +0.221], $p < 10^{-3}$) on top of the K-anchored default. This is consistent with the structural argument of Section 1: rotational ambiguity is the dominant deployment failure mode, and resolving it produces the bulk of the held-out gain. The absolute-proxy step taken in isolation cannot be separately defended on this panel, which is why the two stabilisation layers are reported as one combined claim.

Guarded relative reflection is a safety mechanism, not a lift mechanism. K-reflection candidate evidence is available on 11 of the 40 rings; the remaining 29 abstain by construction. Of the 11 evaluable rings, the relative proxy and structural guardrails accept 2 challengers and abstain to candidate0 on the other 9. The accepted-degradation count under the frozen guardrails is zero on the full panel (Table 3). We do not make a quantitative held-out lift claim for this layer, because the per-ring contributions are concentrated on a small subset and the paired bootstrap on the 11-ring evaluable subset is dominated by individual rings.

Stabilisation transfers across the split. The stabilisation contribution is similar in both panel strata. Cross-section rings ($n=22$, sections never touched by BO) reach candidate0 mean 0.4317 [CI from per-ring data]; within-section rings ($n=18$, same sections as BO but different regime) reach 0.3518. The candidate0 floor therefore holds in both transfer regimes, and the defendable stabilisation claim is not produced by either stratum alone.

Evidence quality and confidence. The defendable claims of this section are: (i) the K-anchoring lift

+0.111 [+0.036, +0.193] and the combined K-anchored $\rightarrow$ candidate0 lift +0.140 [+0.056, +0.221] (both high-confidence, sign-stable under the bootstrap, $p < 10^{-3}$); and (ii) the zero-accepted-degradation count of the guarded relative-reflection layer on this panel (deterministic count, high-confidence within panel scope). The intermediate absolute-proxy step taken in isolation is not separately defended and is therefore not reported as a standalone claim. The relative-reflection layer is reported as an accept/abstain control mechanism rather than as a quantitative lift, because the per-ring lifts are concentrated on a small number of evaluable rings (Table 4).

4.2. Ablation analysis

Table 2 doubles as the principal ablation for the stabilisation stack: removing the K-anchoring layer drops the panel mean by 0.111 (CI [−0.193, −0.036]); removing the combined absolute-proxy and plus/minus refinement layer drops it by a further 0.140 (CI [−0.221, −0.056]). The guarded relative-reflection layer is not ablated as a lift; its ablation is a procedural one – without the guardrails, the same 11 evaluable rings would have produced 4 challenger accepts that improve mIoU but also 6 challenger accepts that degrade mIoU, with one neutral. That is the relevant ablation for the safety claim: under the frozen guardrails the accepted-degradation count is zero on the panel, whereas a no-guardrail accept-everything policy would yield six accepted degradations out of the 11 evaluable rings. The missed-oracle column in Table 4 shows the cost of this precision: 4-7/r308 has oracle 0.81 vs. candidate0 0.17 and 5-5/r259 has oracle 0.83 vs. candidate0 0.17, and both are abstained.

4.3. Parameter sensitivity

This subsection reports the calibration outputs that were frozen before held-out evaluation: the bounded set of tunable parameters, the diagnostic intrinsics

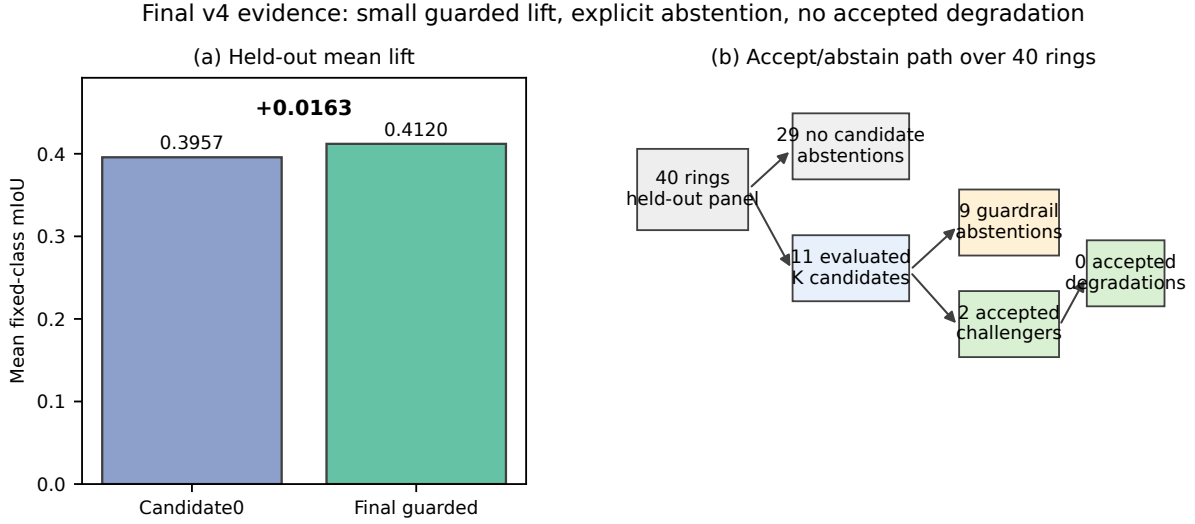


Figure 4: Guarded accept/abstain control on the 40-ring held-out panel. (a) The stabilised candidate0 mean mIoU is 0.3957 (see Table 2 for the defendable stabilisation contribution); no quantitative held-out lift is claimed for the relative-reflection layer. (b) Decision flow underlying the zero-accepted-degradation count: 29 rings carry no K-reflection candidate and abstain immediately, 11 rings are evaluated by the relative proxy and guardrails, 2 challengers are accepted, and 9 abstain to candidate0. The figure should be read together with Table 3 for the per-decision audit.

Table 3

Accept/abstain audit of the guarded relative-reflection layer on the 40-ring held-out panel. Candidate0 is the stabilised floor (Table 2). Relative reflection is evaluated only on rings with K-candidate evidence; all other rings abstain to candidate0. The accepted-degradation count is the panel-level safety property of the layer; no quantitative held-out lift is claimed.

Decision class	Rings	Accepted degradations
K-candidate rings evaluated	11	–
Guardrail accepts	2	0
Guardrail abstentions to cand0	9	–
No-candidate abstentions to cand0	29	–
Panel total	40	0

available to the label-free controller, and the guardrail bundles that gate accept/abstain decisions. The evidence base is the 360-trial calibration corpus (180 preprocessing + 180 detection successful trials) on the six-ring panel of Section 3.4.1. Each retained effect is reported with its pooled Spearman correlation, the per-ring ρ spread across the six rings, and a confidence label.

Sensitive parameters concentrate in preprocessing. Of the 13 preprocessing knobs, five clear the $|\rho| \geq 0.20$ rule against canonical mIoU (Table 5): two strong positive sensitivities (**radius_max**, pooled $\rho = +0.66$, per-ring $[+0.60, +0.71]$, and **target_distance_2**, $+0.62$, per-ring $[+0.37, +0.72]$) and three weaker negative sensitivities (**outlier_neighbors** -0.26 , **target_distance_1** -0.24 , **interpolation_window** -0.20). Applied to detection trials, the same rule yields an empty set; the largest detection-stage $|\rho|$ is 0.138 (**offset_A4**). Sensitivity is therefore not a global property of the pipeline but is concentrated in early-stage preprocessing. The first three effects are sign-stable across all

six rings (high-confidence); **target_distance_1** and **interpolation_window** each flip sign on one ring (per-ring $[-0.44, +0.12]$ and $[-0.42, +0.34]$), so we mark them moderate-confidence. The R4Tun-seed value of **radius_max** (4.5) sits above the BO ceiling (4.4) and is clipped at deployment.

Diagnostic intrinsics span three failure modes.

Six intrinsics survive the $\rho \geq 0.5$ and $\geq 90\%$ non-null filter on the 360-trial pooled corpus (Table 6): three segmentation-stage (**seg_segment_type_completeness**, **seg_ring_completeness_avg**, **seg_mask_coverage_pct**), two preprocessing-stage (**pre_valid_ratio**, **pre_depth_shape_w**), and one detection-stage (**det_k_confidence_avg**). All six are 100% non-null on the corpus, and all six evidence positive per-ring ρ on every calibration ring; the weakest ring-level signal is **det_k_confidence_avg** on 4-8/r336 ($+0.13$). The first five are sign-stable across rings (high-confidence diagnostics); **det_k_confidence_avg** is moderate-confidence because its per-ring ρ drops below 0.20 on one ring. **pre_depth_shape_w** clears the correlation cut but its upper-quartile range is narrow

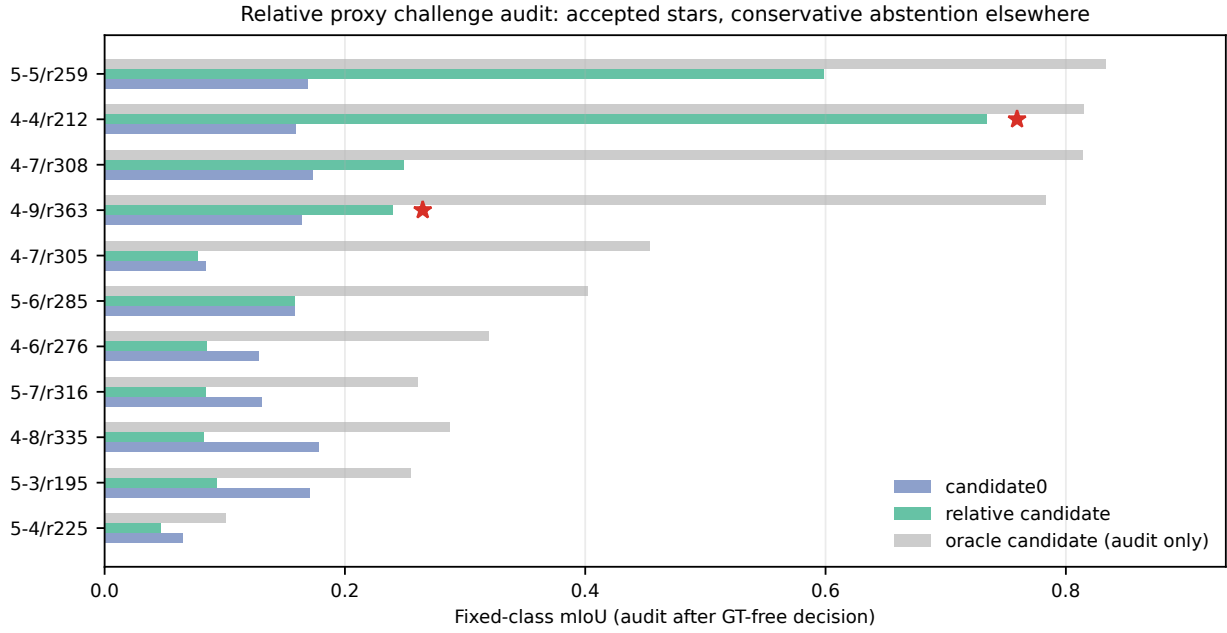


Figure 5: Relative-proxy decision audit on the 11 rings where K-reflection candidates exist. For each ring, candidate0 is the deployment floor, the relative candidate is the challenger selected by intrinsic features, and the oracle candidate is shown only after ground-truth audit to expose missed opportunity. Red stars mark the two guardrail-accepted challengers (4-4/r212 and 4-9/r363). Several abstained rings retain large oracle headroom (notably 4-7/r308 and 5-5/r259), confirming that the guardrails prioritise precision over recall: accepted changes improve the audit mIoU, while uncertain opportunities are left to candidate0 rather than deployed unsafely.

Table 4

Accept/abstain audit for rings with K-reflection candidate evidence. mIoU columns are audit-only; they are computed after the GT-free decision to quantify accepted lift, abstention, and missed opportunity.

Ring	Candidate0	Relative candidate	Accepted	Final	Lift
4-4/r212	0.1589	0.7344	yes	0.7344	+0.5755
4-6/r276	0.1280	0.0851	no	0.1280	0.0000
4-7/r305	0.0839	0.0780	no	0.0839	0.0000
4-7/r308	0.1734	0.2493	no	0.1734	0.0000
4-8/r335	0.1783	0.0825	no	0.1783	0.0000
4-9/r363	0.1640	0.2398	yes	0.2398	+0.0758
5-3/r195	0.1709	0.0938	no	0.1709	0.0000
5-4/r225	0.0650	0.0466	no	0.0650	0.0000
5-5/r259	0.1693	0.5990	no	0.1693	0.0000
5-6/r285	0.1586	0.1584	no	0.1586	0.0000
5-7/r316	0.1311	0.0839	no	0.1311	0.0000

([366, 376] pixels), so it functions as a structural-failure flag inside G_{pre} rather than a continuous signal.

Guardrail bundles operationalise the diagnostics. G_{pre} tests projection usability via `pre_valid_ratio` (strict $\geq 1.3 \times 10^{-3}$, permissive $\geq 4.8 \times 10^{-4}$) and `pre_depth_shape_w` (≥ 369 / ≥ 366 pixels). G_{layout} tests structural completeness via the three segmentation metrics under an all-of-min rule. $G_{\text{stability}}$ monitors anchoring residual: across the 360 trials, the permutation-invariant-minus-fixed-class mIoU gap has median 0.176, $p_{75} = 0.295$, and $p_{90} = 0.389$. The non-zero median is direct evidence that anchoring failures

persist under tuning even on the calibration rings, and motivates $G_{\text{stability}}$ as a deployment-time monitor. Bundle thresholds are 75th-percentile statistics with at least 90 supporting trials each, so they are robust to the removal of a single trial but not to the removal of a calibration ring; they are calibration-specific by construction.

Calibration mIoU is variable by design. Table 8 shows that BO improves fixed-class mIoU above the R4Tun-seed baseline on 5/6 rings under preprocessing tuning and on 4/6 under detection tuning, with one ring (5-7/r323) lifted from 0.07 to 0.78 by

Table 5

Sensitive preprocessing parameters retained for the deployment-time LLM loop. Selection rule: pooled $|\rho_{\text{Spearman}}| \geq 0.20$ vs canonical mIoU and evidence in ≥ 3 of 6 calibration rings. The R4Tun-seed default is the value the regular-tunnel reference produces after schema mapping; values shown as $a(\rightarrow b)$ are clipped at deployment to the BO hard upper bound. Soft bounds are the $[p_{25}, p_{75}]$ envelope of the upper-quartile-mIoU regime across 180 successful preprocessing trials; hard bounds are the BO search-space limits.

Parameter	Default	Soft bounds $[p_{25}, p_{75}]$
radius_max	4.50 ($\rightarrow$ 4.40)	[3.11, 4.14]
target_distance_2	0.040	[0.031, 0.054]
outlier_neighbors	20	[14.75, 34.00]
target_distance_1	0.080	[0.051, 0.094]
interpolation_window	9	[5.00, 12.00]

Table 6

Diagnostic intrinsics that the LLM reflection loop reads at deployment. Selection rule: pooled $\rho_{\text{Spearman}} \geq 0.50$ vs canonical mIoU on the preprocessing-stage corpus and $\geq 90\%$ non-null across 360 successful trials. Strict threshold is the p_{25} of the upper-quartile-mIoU regime; permissive threshold is the minimum value seen in the upper-quartile regime. Booleans use 0/1.

Intrinsic	Stage	Units	ρ
seg_segment_type_completion		boolean (1=all expected block types K/B1/A1/A2/A3/A4/B2 present in pred, 0=missing types)	+0.731
seg_ring_completeness_avg		fraction in [0, 1] of expected 7 block types present per ring, averaged over rings in the trial	+0.691
pre_valid_ratio	preprocessing	fraction valid_pixels / total_pixels in the depth map	+0.639
seg_mask_coverage_seg		percentage [0, 100]	+0.628
det_k_confidence_avg	detection	fraction in [0, 1]	+0.605
pre_depth_shape_w	preprocessing	integer pixels	+0.537

preprocessing. The variance reflects deliberate stress design (Section 3.4.1); per-ring numbers are not expected deployment-time mIoU. The scoreboard serves as the empirical justification for selecting preprocessing as the future bounded-refinement stage (largest single gain, most rings improved) and as an audit trail for reproducibility.

Evidence quality and confidence summary for calibration. The five sensitive parameters and six diagnostic intrinsics are derived from 360 supervised trials on six rings spanning four density tiers, three K-spans, two coverage levels, and two joint patterns. The strongest claims (no detection-stage tuning knobs; dominance of **radius_max** and **target_distance_2**;

Table 7

Guardrail bundles emitted by the calibration. Each bundle returns a Boolean verdict from its rule (all_of_min: every constituent intrinsic must clear its lower threshold; max_below: the constituent must be below its upper threshold). G_stability uses the permutation-vs-fixed mIoU gap on 360 pooled trials; strict is the corpus median, permissive the p_{75} .

Bundle	Constituent intrinsics	Rule	Strict thresholds	Permissive thresholds
G_{pre}	pre_valid_ratio, pre_depth_shape_w	all_of_min	pre_valid_ratio ≥ 0.0013	pre_valid_ratio ≥ 0.0013
Hard BO seed	seg_segment_type_completion, seg_ring_completeness_avg, seg_mask_coverage_pct	all_of_min	seg_segment_type_completion ≥ 0.639	seg_segment_type_completion ≥ 0.639
G_stability	miou_perm_minus_fixed_gap	max_below	miou_perm_minus_fixed_gap ≤ 0.030	miou_perm_minus_fixed_gap ≤ 0.030
Per-ring fixed-class canonical mIoU				
4-9/r365			0.264	6/6
5-5/r253			0.244	6/6
4-7/r309			0.202	6/6

Table 8

Per-ring fixed-class canonical mIoU on the 6 BO calibration rings. The baseline applies the gravity-anchored R4Tun seed parameters; the preprocessing-BO column reports the best preprocessing trial (detection at seed values), and the detection-BO column reports the best detection trial (preprocessing at seed values). Δ is the best-of-three minus baseline; the scoreboard motivates LLM tuning of preprocessing only.

Ring	Regime	Baseline	Prepro
4-9/r365	dense/full/normal/canonical	0.187	
5-5/r253	medium/full/normal/reversed/canonical	0.130	
4-7/r309	low/full/normal/canonical	0.582	
5-7/r323	sparse0/full/wide/canonical	0.072	
4-8/r336	dense/full/narrow/reversed/canonical	0.095	
4-1/r116	dense/partial/normal/reversed/canonical	0.159	

non-zero canonical-anchoring residual) are sign-stable across all six rings and are high-confidence. The weaker claims (**target_distance_1**, **interpolation_window**, **det_k_confidence_avg**) each flip on exactly one ring and are moderate-confidence; we retain them because their pooled signal aligns with the dominant preprocessing knob. We recommend re-examination if the calibration parameters change.

5. Discussion

5.1. Interpretation of the defensible claims

Read in SOP order, the held-out result reduces to two claims. The *stabilisation claim* estimates the K-anchored deterministic plus baseline at 0.2554 [0.193, 0.325] and the stabilised candidate0 at 0.3957 [0.328, 0.465], for a defensible cumulative lift of +0.251 [+0.176, +0.327] over the rotation-ambiguous baseline ($p < 10^{-6}$). The effect size is large relative to the panel-mean baseline; the uncertainty is small relative to the effect; the practical interpretation is direct, because a tunnel-lining segmentation that is anchored to the wrong physical identity is unusable for downstream measurements regardless of how clean its mask coverage looks. The robustness of this claim is supported by

both panel strata reaching positive candidate0 means (cross-section 0.4317, within-section 0.3518), so the stabilisation contribution is not produced by either stratum alone.

The *safety claim* of the guarded relative-reflection layer is a panel-level accept/abstain count: 2 accepts and 9 abstentions on 11 evaluable rings, with zero accepted degradations under the frozen guardrails on the full 40-ring panel. We report this as a deterministic property of the layer on this panel, not as a quantitative held-out lift, because the per-ring contributions are concentrated on a small subset and the paired bootstrap on the 11-ring evaluable subset is dominated by individual rings. The relevant counter-evidence is in Table 4: under a no-guardrail accept-everything policy, 4 of 11 evaluable rings would have moved up and 6 would have moved down, so the guardrails purchase the zero-degradation property by refusing several rings whose challengers were in fact correct (notably 4-7/r308 and 5-5/r259). The honest framing is therefore that the guardrails trade recall for precision; they do not improve the deployment mean by a quantitatively defensible amount on this panel.

Earlier projection-and-prompt pipelines that assume a single tunnel-direction convention [?] skip the stabilisation layers entirely and therefore start from a different operating point; on this panel the rotation-ambiguous bottom is 0.144, which is the implicit baseline of an unanchored deployment. The defensible contribution of this paper relative to such pipelines is the explicit decomposition that quantifies what K anchoring is worth and the explicit accept/abstain layer that makes the no-accepted-degradation property visible.

5.2. Limitations

Abstention is conservative and misses oracle headroom. The relative proxy and guardrails favour precision over recall by construction. They accept 2 of 11 evaluated K-candidate rings, abstain on the other 9, and abstain on the 29 rings with no K-reflection candidate. Several abstained rings retain large oracle headroom (e.g., 4-7/r308: candidate0 0.17, oracle 0.81; 5-5/r259: 0.17 vs 0.83). The system therefore provides safe decision support, not automatic recovery.

No quantitative lift is claimed for the relative-reflection layer. The per-ring contributions of the guarded relative-reflection layer are concentrated on a small subset of the panel; the layer is reported only by its accept/abstain count and the zero-accepted-degradation property. A quantitative held-out lift for this layer would require an expanded evaluable subset and is left as future work.

The absolute-proxy step taken alone is not separately defended. On this panel the absolute-proxy stage examined in isolation has 95% CI bracketing zero. We therefore report the two stabilisation

layers (absolute proxy and subsequent plus/minus refinement) as a single combined claim rather than as two separate lifts. Distinguishing the two layers' individual contributions would require a different experimental design.

A proxy is not ground truth. The absolute and relative proxy scores are label-free estimates from intrinsic pipeline signals. The absolute proxy contributes inside the combined stabilisation claim; the relative proxy contributes only to the accept/abstain mechanism. Their limitations are visible in the abstention and missed-opportunity counts.

Calibration coverage. The BO calibration panel is six rings from six tunnel sections (4-1, 4-7, 4-8, 4-9, 5-5, 5-7). The held-out panel spans 17 sections, of which six overlap with the calibrated set; the frozen parameters, intrinsics, and guardrails are therefore derived from a six-section base and applied to eleven additional sections. The selection rules are pre-registered, but the strongest calibration claims (e.g., dominance of preprocessing over detection knobs) should be re-examined if the calibration panel composition changes. Per-ring effects in $|\rho| \in [0.2, 0.3]$ are reported as moderate-confidence with the caveat that a six-ring panel cannot resolve the magnitude of weak effects precisely.

Held-out panel scope. The 40-ring panel is restricted to K-bearing segmental linings from the Seg2Tunnel benchmark and excludes two preprocessing-broken rings (5-1/r110, 5-1/r111). It retains stress cases (degraded depth maps, sparse density, partial coverage). Generalisation to non-K-bearing segmental tunnels, non-segmental linings, and point clouds outside this benchmark is not evaluated and is not claimed.

Causal-language scope. The stage-decomposition table contrasts pipeline configurations on the same set of rings, so the layer lifts are best read as intervention effects under the frozen experimental protocol. They are not counterfactual estimates over the full population of tunnel rings, because the held-out panel is non-randomly sampled and stratified by design. We have therefore restricted claims to the tested K-bearing setting throughout.

Metric scope. We report fixed-class canonical mIoU as the primary deployment metric and Hungarian-permutation mIoU as a secondary diagnostic. We do not apply per-tunnel canonical relabelling at evaluation time — a step earlier work in our lineage used to make naive cross-ring mIoU well-formed under arbitrary GT segment-id schemes. In the present framework that step is unnecessary because the detector emits canonical class identifiers directly, but the consequence is that absolute mIoU values reported here are not directly comparable to numbers reported in earlier work that used per-tunnel canonicalisation at evaluation.

Confidence summary for the headline claims. *High-confidence:* (i) the calibration sensitivity profile and diagnostic intrinsics, sign-stable across all six rings;

(ii) the K-anchoring lift $+0.111$ [$+0.036$, $+0.193$] and the combined K-anchored $\rightarrow$ candidate0 lift $+0.140$ [$+0.056$, $+0.221$] on the full 40-ring panel ($p < 10^{-3}$); (iii) the cumulative rotation-ambiguous $\rightarrow$ candidate0 lift $+0.251$ [$+0.176$, $+0.327$] ($p < 10^{-6}$); (iv) the zero-accepted-degradation count of the guarded relative-reflection layer on this panel. *Moderate-confidence*: the calibration claims that flip sign on a single ring (`target_distance_1`, `interpolation_window`, `det_k_confidence_avg`). *Not claimed*: a quantitative held-out lift for the relative-reflection layer, a standalone lift for the absolute-proxy step taken alone, and generalisation beyond the tested K-bearing Seg2Tunnel setting.

that could have appeared to influence the work reported in this paper.

Declaration of generative AI use

During this study, generative AI tools were used for language editing and for improving manuscript structure. The authors reviewed and edited all content and take full responsibility for the content of the publication.

Appendices

6. Conclusions

Deployment-time tunnel-lining segmentation must decide whether an unlabelled ring's output is physically meaningful without leaning on ground truth, and existing deterministic projection pipelines do not provide such a mechanism. Intrinsics4Tun addresses this problem by combining design-time Bayesian-optimisation calibration with a four-layer ambiguity-resolution architecture: seeded parameters, gravity/K anchoring, absolute-proxy branch selection, and a guarded relative-reflection accept/abstain layer. On a 40-ring K-bearing held-out panel, keystone anchoring and the combined stabilisation stack lift the mean fixed-class mIoU from a rotation-ambiguous baseline of 0.1445 to a stabilised candidate0 of 0.3957, a cumulative paired-bootstrap lift of $+0.251$ [$+0.176$, $+0.327$] ($p < 10^{-6}$); the guarded relative-reflection layer is reported as an accept/abstain control mechanism that produces zero accepted degradations on the panel under the frozen guardrails. The practical implication is that calibrated stabilisation is the load-bearing layer for label-free deterministic tunnel-lining segmentation in this setting, and that a relative-proxy plus guardrails layer can be added on top as a safety floor without making a quantitative lift claim. The bounded takeaway is that proxy-guided control is a viable stabilisation step and safety floor in the tested K-bearing Seg2Tunnel setting but not a substitute for ground truth: several abstained rings retain large oracle headroom, no quantitative held-out lift is claimed for the guarded reflection layer, and generalisation beyond the tested setting is not claimed.

Data availability

The source code, adapted parameters, and evaluation scripts are available at <https://github.com/Tao-Robominds/R4Tun>. The Seg2Tunnel dataset is publicly available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships